PPR Land Acquisition Under Climate Non-Stationarity

October 26, 2026

Overview

This project asks whether accounting for projected future ecological conditions (non-stationarity) leads to improved land-acquisition decisions for waterfowl conservation in the U.S. Prairie Pothole Region (PPR) than assuming stationarity. It builds three decision models and compares their performance across climate-change scenarios and a stationary null.

The three models acquire land period by period (decadal steps, 2020–2100) under a budget:

- **Greedy** — status-quo heuristic: rank parcels by current benefit-to-cost ratio, buy down the list. No foresight.
- **Myopic ILP** — re-optimizes each decade but freezes current benefit and hazard and assumes they persist (a fabricated stationary future); implements the current decade, then re-solves.
- **Rolling-horizon ILP** — same structure, but uses the true projected trajectories. The foresighted benchmark.

The spatial unit is the **EAU** (Equal Area Unit, ~342 km^2). After dropping 4 Wetland Management Districts (due to either lack of landcover data, or low confidence in hydrological modeling), the analysis uses **879 EAUs across 20 WMDs**, over **9 decadal periods** and **5 scenarios** (RCP 4.5/8.5 × wet/dry, plus a stationary null).

The main (canonical) results use the FOREsce GCAM Reference land-cover scenario and a budget of 5 median-priced EAUs per period. Two sensitivity analyses are built into the workflow: an alternative land-cover scenario (FOREsce BAU) and the size of the budget (2, 5 and 10 median-priced EAUs per period).

How to run

Requires R with the native **Gurobi** R package and an active (academic) license, and Quarto to render the results document. Run everything from the project root.

1. **Main run** — `Rscript 00_run_all.R` runs the data pipeline (`01`–`06`), the models (`08`) and the reporting assets (`10`) for the land-cover scenario set in `00_config.R` (normally `gcam_ref`). Outputs go to `runs/<LC_SCENARIO>/`. Leave the `PPR_BUDGET` environment variable unset for this run.
   - Part of the pipeline can be run with `Rscript 00_run_all.R 08` (from `08` onward) or `Rscript 00_run_all.R 03 06` (a range). `02` is slow, so avoid re-running it unless the land-cover inputs change.
   - `09_validation.R` certifies the formulation (P1–P5). It is skipped by default; set `RUN_VALIDATION <- TRUE` in `00_run_all.R` (or run it directly) when the formulation changes. It does *not* need to be re-run every time `08` is run.
2. **Budget sensitivity** (optional) — `Rscript 13_run_budget_sweep.R` re-runs `08` and `10` at budgets of 2, 5 and 10 median-priced EAUs per period, reusing the derived data from step 1 (land cover is not reprocessed). Each budget gets its own folder, `runs/<LC_SCENARIO>_bXX/`. Then `Rscript 10c_results_assets_budget.R` builds the cross-budget tables and figures. The budget-5 run reproduces the main run, and the sweep checks this. ~30–45 min per budget.
3. **Land-cover comparison** (optional) — set `LC_SCENARIO <- "bau"` in `00_config.R`, run `Rscript 00_run_all.R`, set it back to `"gcam_ref"`, then run `Rscript 10b_results_assets_landcover.R`.
4. **Results document** — `quarto render 11_results.qmd`. The budget (`11c`) and land-cover (`11b`) sections are included automatically when their assets exist.

Input data

`input_data/` is not distributed with the repository. Assemble it as follows:

Path	Source
`input_data/Wetland_Management_Districts/FSMS_WMD.shp` (with `.dbf`, `.prj`, `.shx`, and other sidecar files)	USFWS Wetland Management District boundaries
`input_data/prairie_potholes_gcam_ref_meanclim/` — only `prairie_potholes_gcam_ref_meanclim_2020.tif` is read	USGS FOREsce, GCAM Reference: https://www.sciencebase.gov/catalog/item/5b683db2e4b006a11f75b06a
`input_data/prairie_potholes_gcam_ref_rcp45/` — 10 GeoTIFFs, 2014–2100	as above, RCP 4.5
`input_data/prairie_potholes_gcam_ref_rcp85/` — 10 GeoTIFFs, 2014–2100	as above, RCP 8.5
`input_data/prairie_potholes_bau_{meanclim,rcp45,rcp85}/` — same layout as above	USGS FOREsce, BAU (only needed for the land-cover comparison)
`input_data/PADUS4_1VectorAnalysis_PADUS_Only/PADUS4_1VectorAnalysis_PADUS_Only.gdb`	USGS GAP PAD-US 4.1 Vector Analysis: https://doi.org/10.5066/P96WBCHS
`input_data/benefit.csv`	Waterfowl abundance projections (McKenna et al. 2026)
`input_data/places_fmv_pnas_dryad/1 estimates/places_fmv_vacant.tif`	Nolte 2020 PLACES FMV: https://datadryad.org/dataset/doi:10.5061/dryad.np5hqbzq9

Scripts

Setup and drivers:

- `00_config` — single source of run settings: land-cover scenario (`LC_SCENARIO`), budget (`BUDGET_EAUS_PER_PERIOD`, canonical 5; set from the `PPR_BUDGET` environment variable for budget runs), run IDs and folder paths. No packages are loaded here.
- `00_setup` — loads packages, sources `00_config`, and creates the run folders.
- `00_run_all` — runs the pipeline in order (see How to run). Refuses to re-run `01`–`06` for a budget run.
- `13_run_budget_sweep` — budget sensitivity driver: runs `08` and `10` for each budget, skips budgets already complete, logs each run to `logs/budget_sweep/`, and checks the budget-5 run against the main run. Options: `--dry-run`, `--force`, or budgets as arguments (e.g. `Rscript 13_run_budget_sweep.R 3 7`).

Data pipeline:

1. `01_create_EAUs` — grids the WMD shapefile into EAUs; builds an EAU↔WMD lookup (both in `runs/common/`).
2. `02_EAU_prop_suitable` — proportion of suitable habitat per EAU from USGS FOREsce land-cover projections, over time and by RCP, plus gross habitat loss on unprotected land (PAD-US 4.1, GAP 1–3 masked). (Full resampling run ~6–10 min.)
3. `03_create_data_panel` — assembles the long-format data panel (EAU × year × RCP × GCM), including wet/dry duplication and the stationary scenario. Land cover is joined to EAUs by raster cell (`cell_id`), not by row order.
4. `04_benefit_data` — waterfowl-abundance benefit, interpolated to decadal steps (benefit windows anchored at their midpoints: 2012, 2045, 2085) and distributed within WMDs by suitable habitat. **Drops the 4 excluded WMDs** (→ 879 EAUs).
5. `05_risk_of_loss` — conversion-risk hazard, a gross suitable habitat loss (i.e., habitat erosion proxy) on unprotected habitat, with a small background floor ε.
6. `06_cost_data` — acquisition cost per EAU (Nolte 2020 FMV; 2%/yr inflation).

Data Panel Produced via Data Pipeline:

Variables

Variable	Type	Description
`eau_id`	integer	Unique EAU identifier. Assigned in script 01 by re-indexing valid raster cells 1:N, sorted by WMD (alphabetical) then original cell number. Stable across all scripts.
`wmd_id`	character	Name of the Wetland Management District containing the EAU centroid. 20 distinct values after exclusions.
`year`	integer	Decision time step: 2020 (baseline) and each decade 2030–2100. Nine distinct values.
`rcp`	character	Emissions scenario: `"45"`, `"85"`, `"baseline"` (2020 only), or `"stationary"` (no-change counterfactual).
`gcm`	character	Climate model variant: `"wet"`, `"dry"`, `"baseline"`, or `"stationary"`. Note that `prop_suitable` does not vary by `gcm` — only the benefit data are GCM-specific.
`prop_suitable`	double	Proportion (0–1) of the EAU classified as suitable habitat.
`abs_abundance`	double	WMD-level total duck abundance for that year and scenario, repeated across every EAU in the WMD.
`scaled_abundance`	double	EAU-level share of `abs_abundance`, allocated by habitat area. The benefit term used in the ILP.
`eau_row`	double	Internal row-order index used to align the land cover matrices to the EAU lookup table. `NA` for 2020 and stationary rows. Artifact of construction — not for analytical use.
`total_prop_suitable`	double	Sum of `prop_suitable` across all EAUs within the WMD × year × `rcp` × `gcm` group. Denominator for `habitat_share`.
`habitat_share`	double	`prop_suitable / total_prop_suitable`; the EAU's fraction of its WMD's suitable habitat in that scenario-year. `NA` where the WMD total is 0.
`trans_prob`	double	Per-period risk of habitat loss (transition probability), bounded [0, 1].
`cost`	double	Estimated acquisition cost of the EAU in nominal USD for that decision year. Scenario-invariant.
`cell_id`	integer	Raster cell number in the EAU grid — the permanent by-place key linking each EAU to its location. Used to join land-cover data to the panel by position on the ground rather than by row order. `NA` for 2020 baseline and stationary rows, which are constructed from `eau_id` after the land-cover join and never carry the cell key.

Modeling core:

7. `07_ilp_core` — shared engine sourced by `08` and `09`: scenario-matrix assembly, survival/value/budget functions, the Gurobi solver wrapper, the three policies, and evaluation. Solver settings (a 60 s per-solve time cap, threads, seed, output) and the **spend-down** block (`SPEND_DOWN_MODE`, the value-first/full-budget-deployment tiebreak) live in parameter blocks near the top.
    - Temporal discount factor lives here: defined as 0.95 per decade
    - Spend down mode lives here: Budget deployment is controlled by `SPEND_DOWN_MODE` (default `"spend"` forces the model to exhaust the budget each time period).
    - Budget per period = `BUDGET_EAUS_PER_PERIOD` × the median EAU cost in that period (`make_budget()`). The number of EAUs is set in `00_config`, not here.
8. `08_run_models` — driver: runs the three models on all scenarios, scores each against its true future, and writes results, trajectories, a convergence summary, and the **per-parcel acquisition schedule joined to EAU coordinates** (`acquisition_schedule_spatial.csv`, also embedded in `model_results.rds`). The schedule join reads `runs/common/eau_wmd_lookup.rds`; if absent it warns and writes the schedule without coordinates rather than failing.

- option to run model as "reproducible" lives here. Run single-thread + seeded (`REPRODUCIBLE <- TRUE`) for bit-reproducible reported numbers *and* a stable map footprint.

9. `09_validation` — correctness suite (P1–P5). Formulation tests run with spend-down off; the ordering test runs in the active mode.

Reporting / mapping:

10. `10_results_assets` — single pure-reader reporting script: builds results tables and figures (`ggplot2`/`flextable`) and maps from `08`'s saved CSVs into `output_figs/`. No solver/`07` needed. The maps need only `ggplot2`; `terra`/`sf` are used solely for the optional WMD outline overlay (auto-skipped if absent). Toggle maps with `DRAW_MAPS`.

10b. `10b_results_assets_landcover` — optional reporting script that compares two complete pipeline runs: the base run set in `00_config.R` (normally `gcam_ref`) and an alternative land-cover run (`COMPARE_RUN`, normally `bau`). Reads both runs' `output_data/` results and `derived_data/data_panel.rds`, and writes comparison tables (`cmp_headline.csv`, `cmp_value_added.csv`, `cmp_risk_table.csv`), a conversion-risk figure (`fig_cmp_risk_box`) and `cmp_metadata.json` into the base run's `output_figs/`. Run it only after both pipelines have finished, with `00_config.R` set to the base run. No solver/`07` needed.

10c. `10c_results_assets_budget` — optional reporting script that compares the budget runs made by `13_run_budget_sweep` (every complete `runs/<LC_SCENARIO>_bXX/` folder; each run's budget is read from its saved parameters). Writes tables (`bud_headline.csv`, `bud_value_added.csv`, `bud_convergence.csv`), figures (`fig_bud_improvement`, `fig_bud_value_added`, `fig_bud_overtime`), a map (`map_bud_foresight`) and `bud_metadata.json` into the main run's `output_figs/`. Run it with `PPR_BUDGET` unset. No solver/`07` needed.

11. `11_results.qmd` reports results from `10_results_assets`, with annotation/prose accompanying each figure/graph/map.

11b. `11b_landcover_comparison.qmd` — child document holding the land-cover comparison entry (prose, tables, figure) for the Updates section of `11_results.qmd`. It is included automatically only if `10b_results_assets_landcover` has been run (i.e., `cmp_metadata.json` exists); otherwise the entry is skipped. Not rendered on its own: it relies on objects defined in `11_results.qmd`'s setup chunk.

11c. `11c_budget_sensitivity.qmd` — child document holding the budget sensitivity entry (prose, tables, figures, map) for the Updates section of `11_results.qmd`. Included automatically only if `10c_results_assets_budget` has been run (i.e., `bud_metadata.json` exists). Not rendered on its own, for the same reason as `11b`.

12. `12_compare_runs.qmd` — standalone document comparing the land-cover runs (GCAM reference vs. BAU).

13. `13_run_budget_sweep` — budget sensitivity driver: runs `08` and `10` for each budget, skips budgets already complete, logs each run to `logs/budget_sweep/`, and checks the budget-5 run against the main run. Options: `--dry-run`, `--force`, or budgets as arguments (e.g. `Rscript 13_run_budget_sweep.R 3 7`).

References and sources

Land-cover projections: USGS FOREsce (GCAM Reference / BAU); Sohl, T. L., et al. (2012), *Agriculture, Ecosystems & Environment* 153:1–15. Waterfowl abundance: McKenna, O. P., et al. (2026), *BioScience* biag091. Acquisition cost: Nolte 2020, *PNAS* 117(47):29577 (PLACES FMV raster; Dryad doi:10.5061/dryad.np5hqbzq9). Protected areas: USGS GAP (2024), PAD-US 4.1, https://doi.org/10.5066/P96WBCHS. EAU gridding adapted from a script by Heini Kujala. Prior toy-problem precedent: Rubenstein, Madeleine A., et al. "Cognitive and institutional barriers to effective natural resource management under global change." Conservation Biology (2026): e70393.